

Neelabh Raj Awasthi

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Computer Science undergraduate specializing in backend systems, cloud infrastructure, and applied AI. Hands-on experience building RAG-based document intelligence pipelines, self-supervised world models, and self-hosted Linux server architectures. IBM-certified in Data Science and AI.

TECHNICAL SKILLS

Languages:	Python, C, SQL, Bash
AI / ML / DS:	PyTorch, Scikit-learn, Pandas, NumPy, RAG, Vector Databases (ChromaDB), Self-Supervised Learning, JEPA, Prompt Engineering, Reinforcement Learning
Data Science:	Statistical Modeling, Feature Engineering, Data Visualization (Matplotlib, Seaborn), EDA, ML Pipeline Design
Backend / Web:	Flask, REST APIs, Streamlit, NextCloud
Systems:	Linux (Ubuntu, Debian), Git, Docker (familiar), Cloudflare Tunnels, Edge Networking
Exposure:	C++, Java, Computer Vision, Transformers, Cybersecurity, Arduino

WORK EXPERIENCE

ALBA IT Department <i>AI Engineering Intern</i>	Bahrain May 2025 – Jun 2025
- Architected <i>Fine Lines</i>, an enterprise RAG document-intelligence system using ChromaDB; delivered source-cited policy query responses at sub-second latency. - Enforced multi-tenant security via department-scoped vector databases and role-based access controls, ensuring grounded and auditable retrieval. - Reduced response latency further by integrating the Groq API; added contextual follow-ups and TTS, achieving real-time performance in production.	
Kraya and Kuber, KIIT Student Society <i>Marketing & Data Analytics Intern</i>	Jan 2025 – Apr 2025
Drove data-driven outreach campaigns; tracked engagement metrics to iteratively optimize content strategy.	
KIIT Animal & Environment Welfare Society <i>Outreach & PR Member</i>	Dec 2024 – Mar 2025
Led external communications strategy and managed cross-functional logistics for society events.	

PROJECTS

Latent World Model: Semantic Future-State Prediction	Feb 2026 – Apr 2026
- Implemented a Joint-Embedding Predictive Architecture (JEPA) enabling autonomous agents to anticipate future states in a non-generative latent space. - Built a self-supervised training loop with an EMA-updated Target Encoder to ensure representation stability and prevent dimensional collapse. - Integrated with Gymnasium-based 2D physics environments for latent-space navigation planning; reduced inference latency by 65% vs. generative baselines.	
Dynamic Pricing System for Retail	Dec 2025 – Mar 2026
- Designed ML and rule-based pricing models driven by real-time demand signals and inventory fluctuations. - Benchmarked approaches on real-world adoption criteria, producing a comparative analysis to guide production deployment decisions.	
HR Onboarding Assistant (Amazon Q + RAG)	Aug 2025 – Sep 2025
- Automated HR access-ticket workflows and onboarding communications via Amazon Q integration. - Deployed an internal RAG pipeline for policy query resolution, reducing manual HR lookup time.	
Self-Hosted Cloud & Media Server	Dec 2022 – Apr 2023
- Deployed bare-metal Ubuntu Server with Nextcloud (automated sync) and Jellyfin (high-bitrate streaming) on home hardware. - Bypassed Carrier-Grade NAT using Cloudflare Tunnels with Zero Trust policies and DNS rules; achieved zero-open-port firewall posture with high-availability remote access.	

EDUCATION

B.Tech, Computer Science & Engineering — KIIT University	CGPA: 7.70 / 10 (till Sem 5)
Class XII — Laureate Public School	91.2% · School Topper

CERTIFICATIONS

IBM Data Science Professional Certificate (Coursera) · IBM Introduction to Artificial Intelligence (Coursera) · Git & GitHub Fundamentals for Software Developers (Udemy)